

Topology Midterm

Name:

Student ID

Problem 1 (20):

Problem 2 (20):

Problem 3 (20):

Problem 4 (20):

Problem 5 (20):

Total:

1. (20) Show that X is Hausdorff iff $\Delta = \{(x, x) | x \in X\} \subset X \times X$ is closed.

2. (20) Let X be first and second countable and $f : X \rightarrow Y$ is an open continuous. Show that $f(X)$ is also first and second countable. [Map is open if it sends open sets to open sets.]

3. (20) Let $p : X \rightarrow Y$ be a quotient map. Show that if each $p^{-1}(\{y\})$ is connected and Y is connected then X is connected.

4. (20) Let $p : X \rightarrow Y$ be a quotient map and X is locally connected then Y is locally connected.

5. (20) Show that if Y is compact then the projection $\pi_1 : X \times Y \rightarrow X$ is a closed map.